11 Other minds

Chapter overview

- Examines three accounts of how we ascribe mental states to other people (mindreading)
- Considers an argument for the claim that applying psychological concepts to other people is incoherent
- Introduces the sceptical worry that knowledge of other minds is beyond our reach, and considers two responses to it
- Explores some of the challenges involved in detecting consciousness in non-human animals and AI systems.

Much of waking life is concerned with the mental states of other people. We feel empathy for those who have suffered misfortune, we are angry at those who have insulted us, and we are often puzzled by those who fail to see the world as we do. The capacity to understand the mental states of others – to 'mindread' or 'mentalize', as it is known – lies at the heart of social life. You cannot co-operate with someone without being able to 'read' their mind, nor can you deceive them. No wonder, then, that mindreading has attracted so much philosophical attention.

Mindreading raises three kinds of problems. The first is explanatory: how do we actually do it? What cognitive processes do we employ to identify the mental states of other people? I will call this the psychological problem of other minds, for it is a question about how our minds operate. Although the psychological problem is first and foremost a scientific problem, it does raise a number of distinctively philosophical issues, and philosophers have had a lot to say about it. We will examine some of the main accounts of mindreading in Section 11.1.

The second problem of other minds is conceptual. The practice of mindreading assumes that psychological concepts can be intelligibly applied to other people. However, an intriguing argument calls that assumption into question, for it seems to show that there is something deeply problematic in the very thought that someone else might (say) be in pain. We consider this argument (and possible responses to it) in Section 11.2.

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A third problem raised by other minds is epistemic: under what conditions can we *know* what other beings are thinking or feeling? This is in many ways the central philosophical problem of other minds. As we will see, it takes a number of rather different forms. One focus of discussion concerns our capacity to know the minds of our fellow human beings. Our ordinary mindreading capacities are certainly not infallible, but we do assume that other people's mental states are among the kinds of things that we can know. Sceptics, however, argue that such claims are unjustified. We consider the sceptical challenge of other minds in Section 11.3.

Another aspect of the epistemic problem of other minds concerns our knowledge of other *kinds* of minds. What, if anything, can we know about the mental states of creatures that are very different from us? Section 11.4 focuses on non-human animals, while Section 11.5 addresses the challenge of artificial consciousness. These sections tackle what might be described as the *scientific* problem of other minds, for questions about our knowledge of other kinds of minds are open scientific questions in the way that questions about our knowledge of ordinary human minds are not.

11.1 The psychology of mindreading

Before we consider accounts of the psychology of mindreading, we need to consider what mindreading actually involves, for it is a multifaceted phenomenon. Three aspects of mindreading can be distinguished. First, mindreading requires a grasp of psychological concepts. Clearly, I cannot ascribe pain to you unless I have the concept PAIN. Second, mindreading requires the capacity to accurately apply psychological concepts. A competent mindreader will be able to apply the concept PAIN accurately, distinguishing those individuals who are in pain from those who aren't. Third, mindreading involves the capacity to explain and predict behaviour on the basis of mental state attributions. There is limited utility in being able to tell when someone is in pain if you can't also draw out the implications of this fact for explaining and predicting their behaviour. Some theories attempt to account for all three aspects of mindreading, whereas others are more selective in their focus.

Among the various approaches to mindreading that have been developed, three have been particularly influential.[1] The first treats mindreading as a perceptual capacity. The view here isn't that there is a specific perceptual modality for mindreading, but that the standard sensory modalities – most obviously vision, but perhaps other sensory modalities too – provide us with perceptual awareness of other people's mental states.[2]

The perceptual account is easily motivated, for we certainly *talk* as though we have perceptual access to mental phenomena. We speak of *seeing the joy* on someone's face and of *hearing impatience* in their voice. As Wittgenstein noted, our awareness of other people's states of mind often shares the directness and immediacy that characterizes perception:

> We do not see facial contortions and make inferences from them (like a doctor forming a diagnosis) to joy, grief, boredom. We describe a face immediately as sad, radiant, bored, even when we are unable to give any other description of the features. Grief, one would like to say, is personified on the face.
>
> (Wittgenstein 1953/2009: §225)

Wittgenstein's comments focus on emotional states, but it is tempting to think that mental phenomena of other kinds can also be perceived. For example, it is plausible to suppose that one can simply see that someone is trying to shut a door or pick up a cup.

Despite its attractiveness, the perceptual approach to mindreading faces significant challenges. For one thing, it is incomplete: it provides no account of what is involved in grasping psychological concepts, nor does it provide any insight into our capacity to use our knowledge of mental states to explain and predict behaviour. At best, the perceptual approach explains our capacity to *detect* the mental states of others. But can it do even that?

Some theorists say 'no' on the grounds that mental properties are not observable in the way in which (say) a banana's colour or an apple's shape are. Joy, grief, and boredom might be intimately related to observable properties (such as facial expressions), but they cannot themselves be seen. Strictly speaking, what's observable is only the behavioural manifestation of an emotion and not the emotion itself.

But even if mental phenomena are not themselves observable, it can be argued that knowledge of mental phenomena could nonetheless be perceptual. The thought here is that it might be possible to perceive that an object has a certain property even when the property in question is not observable. As Dretske (1973) once noted, although weight itself is not visible, it is possible to see that someone has lost weight. In a similar vein, one might argue that it is possible to see that someone is in pain even if pains themselves cannot be seen.

A further worry concerns the *scope* of the perceptual account. The claim that certain emotional states, bodily sensations, and intentions are perceivable has some plausibility, for these states are associated with characteristic patterns of bodily responses. Although few of us can articulate what is distinctive of the facial expression associated with (say) worry, there clearly *is* such a facial expression, and we are able to recognize it when we see it (see Figure 11.1).

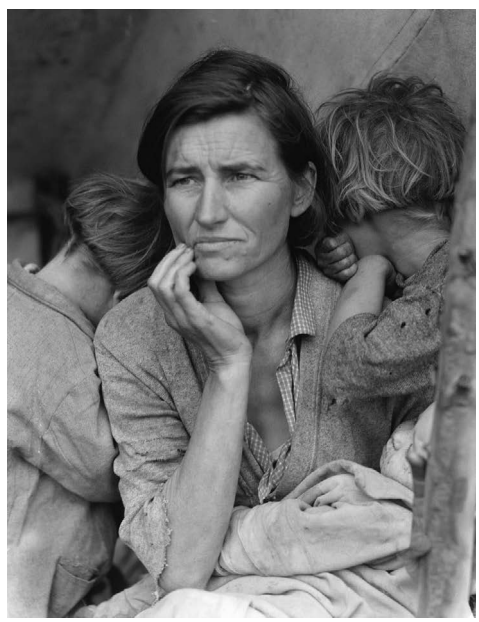

Figure 11.1 Migrant Mother, Dorothea Lange (1936)

But many (perhaps most) mental states are not associated with distinctive patterns of bodily responses. There is no particular way that (say) an intention to learn a foreign language manifests itself in facial expression or bodily movements. In the same way that the holistic and open-ended nature of psychological concepts undermines the plausibility of analytical behaviourism (according to which psychological concepts can be analysed in terms of behavioural dispositions), so too it undermines the idea that mental states are, in general, perceivable. Perception may well play an important role in mindreading, but there is much more to mindreading than perceiving.

Another account of mindreading is the theory-theory, an approach that is so named because its advocates take mindreading to involve the deployment of a kind of folk (or naïve) theory of the mind, often known as 'folk psychology'. The theory-theory has been developed in a number of ways (see Box: The theory-theory: little scientists, mental modules), but at its heart is the claim that we read the minds of others by drawing on a rich body of information about the causal profiles of mental states (i.e., the ways in which mental states interact with each other and with an agent's environment and behaviour). This body of information enables us identify a target's mental states on the basis of their behaviour and environment, and to make predictions about what they will do (or are likely to have done) on the basis of their mental states. For example, if you know that Kylie believes that there is beer in the fridge, and if you know that she wants a beer, then you can use your grasp of the principles of folk psychology ('Other things being equal, if someone desires that X, and if they believe that by doing Y they will secure X, then they will do Y') to predict that Kylie will go to the fridge. Alternatively, if you know that Kylie is going to the fridge, and that she thinks the fridge contains beer, then you might infer that she wants a beer.

Box: The theory-theory: little scientists, mental modules

The theory-theory has been developed in a number of quite different ways. Some theory-theorists hold that children acquire a theory of the mind by using the same general learning processes that are involved in understanding other aspects of the natural and social world (e.g., Gopnik 1993; Gopnik & Wellman 1992). This view is sometimes described as the 'child-as-scientist' account. Other theory-theorists hold that the acquisition of a theory of mind involves the operations of a dedicated mental module, akin to the modules that are implicated in language production and comprehension (e.g., Scholl & Leslie 1999; Baron-Cohen 1995). On this view, our grasp of folk psychology is either innate or is acquired by a specialized learning mechanism. A third version of the theory-theory holds that children acquire a theory of mind from their caregivers and siblings in the course of learning the meanings of psychological terms (e.g., Perner et al. 1994; Peterson & Siegal 2000). The differences between these versions of the theory-theory complicates the task of providing a general assessment of it.

Further reading

Lavelle, J. S. 2018. *The Social Mind*. Routledge.Stich,
S., and S. Nichols. 2003. *Mindreading*. Oxford University Press.

One attractive feature of the theory-theory is that it provides a unified treatment of the three aspects of mindreading that we have distinguished. We have already seen how it accounts for our capacity to both detect mental states and to use information about mental states to explain and predict behaviour. But the theory-theory also provides an account of what is involved in grasping psychological concepts. Roughly speaking, the theory-theory takes psychological concepts to be defined in terms of their functional roles and holds that what it is to grasp a psychological concept is to grasp the relevant functional role.

Critics sometimes object to the theory-theory on the grounds that ordinary people are able to articulate few (if any) of the alleged principles of folk psychology. (Indeed, even philosophers of mind struggle to formulate such principles!) This criticism seems to me to be misguided, for theory-theorists take our grasp of the principles of folk psychology to be 'implicit' or 'tacit' rather than consciously accessible. The model here is the knowledge one has of the grammar of one's native language. Just as the information that explains our capacity to distinguish grammatical sentences ('The girl chased the dog') from ungrammatical ones ('Chased girl dog the') is largely unavailable to conscious awareness, so too – theory-theorists hold – is the body of information that underlies our mindreading capacities.

A third approach to mindreading holds that we work out what mental states another person is in by adopting their perspective, and (roughly speaking) attributing to them the kinds of mental states that we would be in if we were in their situation. This is known as the simulationist account, for it assumes that mindreading involves using one's own mind to simulate the mental states of another person.[3] The idea of simulation-based knowledge is a familiar one. Suppose that you want to know how an Airbus A380 aeroplane will behave in a hurricane. Instead of applying a physical theory to this problem, one might simply build a model of an A380 and see how it behaves when exposed to hurricane-like conditions. So too, simulationists argue, one can track the mental states of another person – the mindreading 'target' – by using one's own mind as a model of theirs. Crucially, simulation-based mindreading doesn't require a theory of how minds work.[4]

Here's an example of how simulation-based mindreading might work. Suppose that we are playing chess and I am trying to predict your next moves. Rather than draw on a theory of people's chess-playing behaviour, I might simply pretend to have your aims and beliefs, and then feed those states into my own decision-making processes. ('Suppose that I were Black. What would I do? Let's see. I could take White's bishop, but then . . .'). If the upshot of this simulation is a decision to move my queen, then I will ascribe that decision to you. More generally, simulationists hold that mindreading involves adopting the target's position, letting one's own psychology run 'offline' on the mental states that are generated by the pretend states, and then ascribing the resulting mental states to the target.

Of course, not all minds are alike. We differ from each other both in our specific mental states (you might have beliefs and desires that I don't) and in our psychological capacities (you might be better at chess than I am). But if our minds differ, how then am I able to use my mind to model yours?

In response, simulationists argue that our mindreading efforts are successful only to the extent that we are able to compensate for such differences. Suppose that you are playing chess with Magnus Carlsen, the current world champion. Although you can pretend to have his aims, you probably won't be able to predict his next move. But the fact that simulation

wouldn't be able to support effective mindreading in this context is actually a mark in favour of the simulationist acount, for this is precisely the kind of situation in which most of us would fail to be effective mindreaders.

Another objection is that merely pretending to have someone else's perspective can't do the kind of work that the simulationist account requires. After all, imagining being in a *physical* state rarely has the effects associated with actually being in the relevant state. For example, drinking hemlock causes death, but merely imagining that one has drunk hemlock has no adverse effects on one's health whatsoever. Thus, critics ask, why should we assume that 'pretend' mental states will trigger the same kinds of psychological processes that are generated by their genuine counterparts?

This is a good question, and providing a comprehensive response to it goes beyond the scope of this chapter. That said, it is clear that there is significant overlap between the causal roles of genuine mental states and that of their imagined counterparts. Horror films generate feelings of fear and heart palpitations despite the fact that one doesn't really believe that one is in danger. One explanation for the commonalities in causal roles between genuine mental states and their imagined counterparts involves the fact that they have content in common. For example, believing that one is facing a grizzly bear is likely to elicit the thought that one is in danger, while pretending that one is facing a grizzly is likely to elicit the (pretend) thought that one is in danger.

We have outlined the three main accounts of mindreading, but we have barely scratched the surface of the debate between these views. In particular, we have not considered the extensive body of research in developmental psychology, psychopathology, and neuroscience which bears on this issue (see, e.g., Apperly 2010; Baron-Cohen et al. 2013; Koster-Hale & Saxe 2013; Lavelle 2018). Unfortunately, we must leave that fascinating work to one side, and turn instead to an argument that challenges the very intelligibility of third-person mindreading.

11.2 The conceptual problem of other minds

Mindreading is woven into the fabric of everyday life, and there would seem to be nothing more natural than applying psychological concepts to other people. There is, however, an influential argument for the claim that it is incoherent to ascribe psychological concepts to other people. This problem is known as the conceptual problem of other minds.[5]

The conceptual problem begins with the idea that grasping a psychological concept requires introspective attention to mental states of the relevant kind. On this view, grasping the concept PAIN (for example) involves attending to one's own pains, and noting that they share a distinctive phenomenal quality (painfulness). This account of what is involved in grasping psychological concepts is attractive and widely endorsed. (Indeed, we saw in Chapter 8 that it lies at the heart of the phenomenal concept response to the knowledge argument.) But – the argument continues – if it's right, then it would be impossible to conceive of someone else's pains. As Wittgenstein put it, 'If one has to imagine someone else's pain on the model of one's own, this is none too easy a thing to do: for I have to imagine pain which I *do not feel* on the model of the pain which I *do* feel' (1953/2009: §302). The problem, in a nutshell, is not that I can't *know* whether you're in pain, but that I can't even formulate the hypothesis that you might be in pain, for my

concept PAIN can be applied only to my own experiences. I can conceive of a pain that is *located* in your body, but I would – so the argument goes – have to think of this pain as mine rather than yours.

We can put the argument as follows:

1 Grasping the concept PAIN is a matter of grasping how pains feel.
2 Given (1), for me to conceive of pain in your body is for me to conceive of a state, located in your body, that feels a certain way.
3 But if a state feels a certain way, then it must feel a certain way *to me*.
4 But a state that feels a certain way to me must be my pain.

Thus,

C it is not possible to intelligibly apply the concept PAIN to someone else.

Since we clearly can intelligibly apply the concept PAIN to other people, there must be something wrong with the argument, but where exactly does it go wrong?

Wittgenstein rejects the first premise. On his account, pains are not 'inner states', and the concept PAIN is not acquired by attending to the phenomenology of one's own pains. Although Wittgenstein doesn't explicitly deny the existence of inner states (feelings of pain), the point of his famous 'beetle in the box' thought experiment is to undermine the idea that grasping psychological concepts involves attending to inner states (see Box: The beetle in the box).

Box: The beetle in the box

One of the most delightful thought experiments in the whole of philosophy is Wittgenstein's famous 'beetle in the box' scenario (1953/2009: §293). Suppose, Wittgenstein says, that everyone has a box into which only they can peer, and that each box contains an object that the owner of the box refers to as a 'beetle'. Because no one has access to the contents of anyone else's box, it is impossible to tell whether any two boxes contain objects of the same kind. Wittgenstein's point is that if terms like 'pain' referred to sensations that were accessible to only one person (in the way that each beetle is accessible to only one person), then there would be a real question about whether different people mean the same thing by 'pain'. But, Wittgenstein continues, since we clearly *do* mean the same thing by 'pain' (and other psychological terms), it follows that 'pain' (and psychological terms more generally) cannot be the name of a private object (a 'beetle in a box'). Wittgenstein concludes that psychological terms acquire meaning on the basis of their links to publicly accessible behaviour rather than on the basis of their relations to inner states of consciousness.

Further reading

Budd, M. 1989. *Wittgenstein's Philosophy of Psychology*. Routledge.
Wittgenstein, L. 1953/2009. *Philosophical Investigations*. 4th ed. P.M.S. Hacker and Joachim Schulte (eds. and trans.). Oxford: Wiley-Blackwell.

Although theory-theorists distance themselves from what they regard as the overly behaviouristic flavour of Wittgenstein's account, they share his rejection of (1). Understanding the concept PAIN, they hold, isn't a matter of grasping how pains *feel* but of understanding the role that the concept of pain plays within folk-psychological theory – roughly, that is, understanding what causes pain and what pains in turn cause.

Many theorists, however, will have qualms about rejecting (1). Although it's certainly plausible to suppose that *some* psychological concepts can be grasped in the absence of introspective engagement with their objects, that doesn't seem to be true of the concept PAIN. Arguably, grasping PAIN requires knowing what it's like to be in pain. Someone who has never experienced pain might be able to acquire a *kind* of concept of pain, but – the thought is – they won't have a full grasp of the ordinary concept of pain. Thus, it is a matter of some interest whether the conclusion of the conceptual argument can be resisted without rejecting (1).

The answer, I think, is that it can. Premise (3) is surely implausible: a state can feel a certain way without feeling that way *to me*. Even if the concept PAIN can be acquired only via introspective attention to one's own pains, it doesn't follow that one cannot intelligibly apply it to other people. The key here is the distinction between one's own subjectivity and subjectivity as such. It may indeed be incoherent to suppose that a pain can feel a certain way without feeling a certain way *to someone*, but it doesn't follow that the person feeling the pain must be identical to the person deploying the concept PAIN. The idea that we grasp psychological concepts by attending to our own mental states might indeed be false, but the fact that it is intelligible to apply psychological concepts to other people doesn't show that it is false.

11.3 The sceptical problem of other minds

We ordinarily assume that the individuals with whom we interact on a daily basis have minds, and that we know quite a bit about their mental lives. The sceptic, however, argues that these assumptions are unjustified, and that we know little (if anything) about the minds of others. Mindreading, according to the sceptic, is not a source of knowledge. Indeed, the radical sceptic argues that we don't even have the capacity to form warranted beliefs about other minds.

There are two ways of engaging with the sceptical problem of other minds. The first aims to prove the sceptic wrong, and to demonstrate to the sceptic's satisfaction that our ordinary mindreading methods deliver knowledge. In my view, this position is misguided, for it seems to me that we are justified in assuming that the sceptic is wrong (Box: Solipsism). To my mind, a better way of engaging with the sceptical problem is to treat it as issuing a challenge regarding the nature of our knowledge of other minds. The aim, then, is not to prove *that* we have knowledge of the minds of our follow human beings, but to identify the *basis* of this knowledge. With that thought in mind, let's consider what the three accounts of mindreading that we canvassed in Section 11.2 (the perceptual theory, the theory-theory, and the simulation-theory) have to say with respect to the sceptical challenge.[6]

Box: Solipsism

Most philosophers assume that there are other minds, and that we can (and do) know quite a bit about them. Solipsists reject these assumptions, for the solipsist thinks of herself/himself as the only conscious subject that there is. The solipsistic perspective is eloquently captured by Sylvia Plath in the opening verse of her poem *Soliloquy of the Solipsist*:

> I?
> I walk alone;
> The midnight street
> Spins itself from under my feet;
> When my eyes shut
> These dreaming houses all snuff out;
> Through a whim of mine
> Over gables the moon's celestial onion
> Hangs high.

Solipsism is a vanishingly rare view – indeed, there is a sense in which no two solipsists hold the same view, for each denies the other's existence. However, the threat of solipsism frames the sceptical challenge of other minds, for it is often held that an adequate response to the sceptical challenge must show that solipsism is unreasonable.

Plath, S. 1981. Soliloquy of the Solipsist. *Collected Poems*. Harper Perennial.

Let us begin with the perceptual account of mindreading. If mindreading is perceptual, then its epistemic credentials would seem to be secure, for perception is surely a source of knowledge. Just as my belief that there is (say) a barn in front of me is justified by the fact that it looks like there is a barn in front of me, so too – one might argue – my belief that you are angry is justified by the fact that you look angry. Of course, perceptually based beliefs *can* turn out to be false (perhaps what looks to be a barn is just a barn façade; perhaps you are merely pretending to be angry), but to acknowledge that perception can mislead is not to say that it ordinarily misleads or that it is typically unwarranted. At the very least, if our access to other minds is perceptual, then it is no worse off than our access to many other features of the external world.

It is, however, very much an open question whether our primary mode of access to other minds *is* perceptual. As we noted earlier, although we may have perceptual access to certain kinds of mental states (such as emotions), it is unlikely that mindreading is in general perceptual. Thus, any response to the sceptical challenge that appealed only to perception would be severely impoverished indeed.

What about the simulationist account of mindreading? Would mindreading generate knowledge if it were primarily a matter of simulating another person's perspective? The simulationist account of mindreading goes hand in glove with what is known as the

analogical account, a view that is most closely associated with the writings of J. S. Mill (1865/1979) and Bertrand Russell (1948).[7] The idea is this: because groaning and wincing are associated with pain in my case, I can infer that you are in pain from the fact that you are groaning and wincing. More generally, the analogical response holds that mind-reading generates knowledge because it is reasonable to suppose that the correlations between mental states and behaviour which apply in one's own case will apply also to other people.

Let's consider three objections to the analogical account. The first derives from the fact that it is relatively rare for behaviours to be uniquely associated with particular mental states. Suppose that you see me run from my house. I might be running because: I believe that my house is on fire and I want to escape from it; I am going to the aid of a neighbour; or because I am exercising. But if a single type of behaviour can be associated with indefinitely many mental states, how could inferences from behaviour to mentality ever be justified (let alone underpin claims to knowledge)?

The response to this objection involves recognizing that the interpretation of behaviour needn't rest on a single, narrowly defined item of behaviour. Instead, we interpret behaviour on the basis of the target's overall behavioural profile. If that profile is consistent with a number of mindreading interpretations (as it often is), then it may be the case that no single interpretation of the target's behaviour will yield knowledge.

A second objection is that the analogical account cannot deliver knowledge because it involves one-shot induction – in other words, it's an inference from a single case. But – the worry runs – one-shot inductions don't underwrite knowledge. Suppose that you suffer from migraines after eating blueberries. Would that give you good reason to think that I too suffer from migraines after eating blueberries? Surely not. You might, of course, happen to be right if you were to draw that inference, but you surely wouldn't *know* that I have a migraine merely because you've seen me eat blueberries. But – the sceptic continues – the blueberry argument (as we can call it) is no worse than ordinary ascriptions of mentality as the analogical account conceives of them. Put another way, the problem is to explain why one-shot induction is legitimate in the pain/wincing case but illegitimate in the migraine/blueberries case.

There is a serious challenge here, but it needs to be carefully formulated. The problem (as I see it) isn't with one-shot induction as such, for one-shot induction *can* yield knowledge. Suppose that you have examined only one female wombat and have observed her give birth to live young. On the basis of this one case, it would be reasonable to conclude that female wombats in general give birth to live young. Whether or not a one-shot induction is warranted depends on how the property in question is distributed in the relevant population. One-shot induction can generate knowledge when dealing with properties that are highly correlated (being female / giving birth to live young) in a particular population (say, wombats), but it can lead one astray when dealing with properties that are not highly correlated (e.g., eating blueberries / having migraines) in the relevant population.

At this point, there are various strategies open to the sceptic. One strategy would be to argue that we have positive reason to doubt that mental properties and behavioural properties are highly correlated. The problem with this strategy is that it requires knowledge of

other minds, which is precisely what the sceptic claims we lack. A less ambitious sceptical strategy would be to argue that we have no good reason to think that mental properties are highly correlated with behavioural properties, and that without such a reason it would be sheer hubris to employ one-shot induction in this domain.

This strategy is more promising, but it too faces objections. For one thing, it is reasonable to think that there are robust connections between mental states and certain patterns of behaviour. One needn't think that these connections are discernible a priori or that they capture the essence of mental phenomena. All that is required, rather, is that the relations between mental states on the one hand and behavioural responses on the other are relatively stable across individuals. This reply can be motivated by noting that mental properties are evolved properties, and selection pressures are likely to ensure that there is at least broad uniformity in how they are distributed within a species (Sober 2000).

Of course, as we move away from the typical members of our own species and consider the members of other species (and atypical members of our own species), we have less reason to think that one-shot induction will yield knowledge. But this is not a mark against the analogical account, for that account is concerned with *ordinary* mindreading, and, for the most part, ordinary mindreading is restricted to neurotypical, adult members of our own species. After all, we often *don't* know what to say about the mental states of non-human animals, infants, or our fellow human beings who have suffered severe brain damage (see Box: Detecting consciousness in the 'vegetative state').

Box: Detecting consciousness in the 'vegetative state'

Patients with severe brain damage often enter a state of unresponsive wakefulness, known as the 'vegetative state' or the 'unresponsive wakefulness syndrome'. These patients show signs of wakefulness – they exhibit a sleep/wake cycle and spontaneously open their eyes – but they do not produce any behavioural indicators of awareness (hence the term 'vegetative'). However, recent brain-imaging and EEG studies indicate that as many as 20% of these patients may retain some form of awareness (Edlow et al. 2020; Kondziella et al. 2016). The first study of this kind involved a 23-year-old woman who was in a vegetative state as the result of a car accident (Owen et al. 2006). On some trials she was instructed to imagine playing tennis, and on other trials she was instructed to imagine walking around her house. Astonishingly, neuroimaging revealed brain activity in just those brain areas that are known to be involved in motor imagery and spatial navigation. Even more impressive was the temporal profile of this neural activity: it began immediately after the imagery instruction was given and ceased immediately after she was told to stop imagining herself playing tennis or walking around her house.

The advocate of the analogical account might argue that this patient was conscious on the grounds that she displayed the kinds of brain activity that is correlated with consciousness in neurotypical individuals. The advocate of the inference to the best explanation (IBE) account (see below) might argue for the same conclusion on the

grounds that ascribing consciousness to this patient provides the best explanation of her brain activity. Whatever the fate of these arguments, it is clear that the problem of other minds has profound moral and clinical significance.

Further reading

Fernández-Espejo, D., and A. M. Owen. 2013. Detecting Awareness after Severe Brain Injury. *Nature Reviews Neuroscience*, 14: 801–9.
Owen, A. M. et al. 2006. Detecting Awareness in the Vegetative State. *Science*, 313: 1402.
Shea, N., and T. Bayne. 2010. The Vegetative State and the Science of Consciousness. *British Journal for the Philosophy of Science*, 61: 459–84.

What might the sceptical challenge look like from the perspective of the theory-theory? Recall that the theory-theory takes mindreading to involve the application of a theory of mind (folk psychology), which explains behaviour in terms of interactions between an agent's environment and a range of psychological states (perception, beliefs, desires, intentions, etc.). From this perspective, the sceptical challenge can be met by showing that the explanations of behaviour provided by folk psychology are good explanations (or at least, are better than their competitors). This response to the sceptical challenge is known as the *inference to the best explanation* account (Melnyk 1994; Pargetter 1984).

Inference to the best explanation (IBE) is a form of explanation in which one argues for the truth of a hypothesis on the grounds that it provides the best explanation for a certain phenomenon (Lipton 2004). For example, one might defend the hypothesis that it rained last night ('H') on the grounds that this hypothesis provides the best explanation of the fact that the streets are wet. Other explanations are possible (perhaps a water main burst), but if they are weaker (all things considered) than H, then it's reasonable to adopt H (and one might count as knowing H). Similarly, the advocate of the IBE approach argues that we are justified in ascribing folk-psychological states to other people on the grounds that doing so provides the best explanation of their behaviour. Other explanations may be possible, but they will be inferior to those provided by folk psychology.

There are two ways of pushing back against the IBE account. First, one might accept that the ascription of mental states is in the business of explaining behaviour but argue that it is not particularly successful in this regard. As we saw in Chapter 3, eliminativists such as Paul Churchland argue that many of the explanations provided by folk psychology are poor and will be displaced by those that appeal to neuroscience and scientific psychology. Here is not the place for a detailed evaluation of the eliminativist challenge, but it is important to recognize that in some sense there are as many varieties of eliminativism as there are components of folk psychology. Even if certain facets of folk psychology are vulnerable to elimination, others seem to be relatively immune to the eliminativist threat. My own view is that the explanatory credentials of many of the central folk-psychological categories (most notably belief, desire, and intention) are secure.

A second challenge to the IBE account concerns mental phenomena that don't appear to be in the business of explaining behaviour, such as states that are characterized in

terms of their phenomenal character. Consider 'inverted spectra' scenarios of the kind that we discussed in Chapter 4, in which properties that produce one kind of experience in me (say, that associated with the colour of ripe strawberries) produce another kind of experience in you (say, that associated with the colour of broccoli). The possibility of spectrum inversion seems to be irrelevant as far as predicting and explaining your behaviour is concerned. When it comes to figuring out what you're going to do, what matters is not the phenomenal character of your experiences but whether your representation of a class of objects is stable across changes in context. To put the point in Wittgensteinian terms, it doesn't matter whether the beetles in our respective boxes have the same intrinsic nature; instead, all that matters is that they have the same causal profile. As long as we agree on a sufficiently wide range of judgments ('this one is red; that one is green'), it doesn't matter whether the things that look red to me look the way that green things look to you or vice versa.[8]

What should we say then about our knowledge of the phenomenal states of others? Arguably, we have two options. On the one hand, we could embrace the analogical account. Even if the analogical account cannot justify our claim to *know* the phenomenal states of other people, it might be able to provide our beliefs about this domain with a rational foundation. Alternatively, we could reject the 'raw feels' account of phenomenal states, and argue that phenomenal properties are really just intentional/functional states of a certain kind (and as such can, perhaps, be accommodated within the IBE framework). Whether phenomenal states can be reduced to intentional/functional states is, of course, another question.

11.4 Other kinds of minds I: non-human animals

A very different version of the epistemic problem of other minds concerns minds that are unlike ours. When it comes to our fellow human beings, we generally assume that our ordinary mindreading practice can (and often does) generate knowledge. (At least, this is true of adult neurotypical humans: ordinary mindreading struggles to identify the mental states of infants and individuals suffering from severe brain damage.) By contrast, it is very much an open question whether knowledge of the minds of non-human animals and AI systems *is* available to us. At the very least, it is not something that we take for granted. Genuine doubt about whether your friends and family members are conscious is a sign of psychiatric illness, but uncertainty about consciousness in birds, bees, or Data, the android depicted in *Star Trek: The Next Generation* is – one might argue – the appropriate attitude to adopt. Let's postpone for now the problem of artificial consciousness, and consider some of the many challenges posed by consciousness in non-human animals.

One challenge concerns the distribution of consciousness. Is consciousness restricted to (say) mammals, or does it also include reptiles, birds, or fish? What about invertebrates such as bees, lobsters, and spiders? Indeed, might the circle of sentience even encompass plants? (See Box: Plant mentality.) Many of the issues raised by these questions can be brought into focus by considering one particularly intriguing animal: the octopus.[9]

Box: Plant mentality

Botanists often describe plants as perceiving, learning, and even communicating with each other. Most theorists view such talk as metaphorical, arguing that plants no more try to grow towards the sun or tell each other about the presence of pests than laptops try to print documents or smartphones tell their owners about upcoming appointments. Others, however, hold that we should take talk of plant mentality quite literally, arguing that plants perceive, learn, and communicate in much the same sense in which you and I do. In defence of this view, they point out that plants respond adaptively to their environment, they integrate information from different sources, and they track the spatial and temporal gradient trajectories of resource availability. Indeed, some theorists argue that we should even take seriously the idea that plants are conscious – that they have a subjective point of view (Calvo 2017; Gagliano 2017).

Given our ignorance of the physical basis of consciousness, it would surely be premature to dismiss the possibility of consciousness in plants. (Certainly those who take panpsychism seriously have every reason to give the hypothesis of plant consciousness a sympathetic hearing!) At the same time, the case for admitting plants to the consciousness club is far from overwhelming. Rather than think of plants as having an experiential perspective in the way that animals do, it may be more plausible to compare them to the kinds of subpersonal information-processing systems involved in interpreting speech, returning a tennis serve, and picking up a cup of coffee. These systems do indeed have representational states, but there is little reason to treat them as conscious agents in their own right.

Further reading

Calvo, P. 2017. What Is It Like to Be a Plant? *Journal of Consciousness Studies*, 24(9–10): 205–27.
Gagliano, M. 2017. The Mind of Plants: Thinking the Unthinkable. *Communicative and Integrative Biology*, 10(2): e1288333.
Maher, C. 2017. *Plant Minds: A Philosophical Defense*. Routledge.

Octopuses have relatively large brains – the common octopus has roughly 500 million neurons – and they display many features of general intelligence. For example, they have been found using their ability to generate jets of water from their arms to clean out sites to use as a home, repel scavenging fish and pesky experimenters, and short out the lighting in their aquarium that appears to be bothering them (Mather 2019; Godfrey-Smith 2016). Behavioural complexity of this kind in a mammal would generally be regarded as good evidence of consciousness, and it is tempting to regard it as evidence of consciousness in the octopus.

At the same time, there are important differences between us and the octopus, some of which might undermine the plausibility of ascribing consciousness to the octopus. One consideration is evolutionary: the octopus belongs to a very different branch of the tree of life from

that on which we (and other mammals) are located. Thus, either our last common ancestor – which was likely some kind of a small, flattened worm that lived roughly 600 million years ago (Godfrey-Smith 2013) – was also conscious, or consciousness has evolved more than once. Although one (or even both) of these hypotheses might be true, the point is that there are evolutionary considerations against ascribing consciousness to an octopus that don't apply to mammals who display comparable levels of cognitive and behavioural complexity.

Another difference between us and the octopus concerns the structure of cognitive control. Like all cephalopods, the octopus has a decentralized nervous system, and only two-fifths of its neurons are located in its brain; the remainder are in its arms. This distributed control structure appears to provide each of the octopus's arms with significant degrees of autonomy. There is evidence that each arm can process information about its environment without needing to send signals to the brain, and much of what the octopus does seems to be under the control of one or more of its arms acting of its/their own accord. (Hochner et al. 2006).

What implications does this distributed structure of octopus cognition have for the question of consciousness? On the one hand, one could argue that consciousness is necessarily unified, and that an organism is conscious only if it has a single, integrated, point of view on its environment. Those tempted by this response will see the distributed nature of control and perceptual processing in the octopus as evidence that it is a zombie – smart, perhaps, but not conscious. Another view (and one with which I'm more sympathetic) is that the distributed nature of octopus cognition doesn't undermine the case for ascribing consciousness to the octopus, but merely suggests that what it's like to be an octopus is surely very different from what it's like to be a human being (Carls-Diamante 2017). The problem we confront here is that of determining whether a particular feature of human consciousness (in this case, its unity) is essential to consciousness as such, or whether it merely characterizes the form of consciousness that we (together, perhaps, with the members of some other species) enjoy.

Let's turn now from consciousness as such to the *contents* of consciousness. Assuming that we can tell whether a particular kind of animal is conscious, it's a further question whether we can also tell what kinds of conscious states that creature is capable of. A particularly important aspect of this question concerns *self*-consciousness: do other species have the capacity for self-consciousness, or is self-consciousness a uniquely human phenomenon?

Investigations of self-consciousness in humans often focus on mastery of the first-person pronoun ('I'), but obviously that approach cannot be employed when dealing with non-linguistic creatures. Instead, many experimenters employ the mirror test. Developed by Gordon Gallup in the 1970s (Gallup 1970; Gallup et al. 2002), the mirror test involves placing a mark on an animal's head that it can see only with the use of a mirror. An animal is taken to have passed the mirror test when it uses the mirror to engage in self-regarding behaviour (such as attempting to rub the mark off). Chimpanzees pass the mirror test, as do most great apes (other than gorillas); monkeys, by contrast, do not. There is suggestive evidence of mirrored-self recognition in elephants, dolphins, magpies, and even fish.[10]

Is the mirror test a good test for self-consciousness? Failing to pass the mirror test certainly doesn't show that an animal isn't self-conscious, for there are many reasons why an animal could be conscious of itself but refrain from using mirrors to engage in self-directed behaviours. (For example, it might not be interested in removing a mark from its face, or it might find mirrors aversive.) Instead, the question is whether passing the mirror test is

good evidence of self-consciousness. More specifically, we need to ask what *kind* of self-consciousness might be involved in mirrored-self recognition.

The puzzle here is that most animals are clearly aware of themselves in some sense, for they treat their own body parts very differently from the ways in which they treat the body parts of other animals. In this sense, then, even animals that fail the mirror test can be said to have some kind of self-awareness. What does the use of mirrors to inspect one's body add to this? Why should using a mirror to remove a mark from a body part that can be seen only with a mirror indicate a more sophisticated form of self-consciousness than attempting to remove a mark from a body part that can be seen without the aid of a mirror?

One plausible answer is that passing the mirror test requires appreciating how one's body appears from other points of view, and that it is one object among other objects. This is a more sophisticated form of self-consciousness than the kind of self-consciousness that is involved in the fact that an animal will treat its own limbs differently from the way in which it treats the limbs of its conspecifics.

Yet another form of self-consciousness involves what psychologists refer to as *mental time travel* (MTT; Corballis & Suddendorf 2007). This term refers to the ability that you and I have to remember the past and anticipate the future 'from the inside'. For example, I can recall visiting Darjeeling as a child, and I can also anticipate returning to Darjeeling in the future.

MTT is a distinctively human trait, but is it also a *uniquely* human trait? Do other species also have the capacity to remember the past and anticipate the future from the inside? The psychologists Nicola Clayton and Anthony Dickinson (1998) attempted to answer this question by giving scrub jays worms and peanuts to cache in different locations. Scrub jays prefer fresh worms to peanuts, but they prefer peanuts to worms that have been cached for some time (and may have become inedible). Clayton and Dickinson found that the scrub jays retrieved freshly cached worms before they retrieved peanuts, but they retrieved peanuts in preference to worms that weren't fresh.

It is tempting to take these results to show that scrub jays engage in MTT. The thought here is that they attempt to retrieve freshly cached worms first because the anticipated pleasure of eating fresh worms is greatest; and they are able to retrieve the freshly cached worms because they are guided by their experiential memories of when and where they stored the worms. Another possibility, however, is that although the behaviour of scrub jays is guided by their preferences and propositional knowledge, none of that information involves MTT. According to this account, the preference that scrub jays have for fresh worms over peanuts does not involve anticipating what it would be like eat a fresh worm as opposed to a peanut, but is simply a matter of stored information to the effect that fresh worms are tastier than peanuts. Similarly, the propositional account holds that although the behaviour of the scrub jays is guided by information about when and where they cached certain foods, scrub jays don't remember their acts of caching from the inside in the way that you and I might.[11]

It is not clear how we might decide between these two interpretations of the scrub jays' behaviour. The propositional interpretation might be defended on the grounds that it is apparently more parsimonious, requiring only mechanisms of preference formation and propositional memory which we already know non-animals have. At the same time, the MTT interpretation might be defended on the grounds that *we* would employ MTT to execute tasks

like this, and that the methods which we would use are unlikely to be unique to us. The overarching challenge here, of course, is to identify experimental methods that might enable us to distinguish the experiential aspects of 'mental time travel' from its informational aspects in non-verbal creatures.[12]

11.5 Other kinds of minds II: artificial consciousness

As challenging as the problems of consciousness in non-human animals are, it is questions about consciousness in AI systems that threaten to pose the most serious version of the problem of other minds. From Ridley Scott's *Blade Runner* to Alex Garland's *Ex Machina* and Kazuo Ishiguro's *Klara and the Sun*, questions about artificial consciousness have long fascinated fans of science fiction, but with the increasing sophistication of AI they are now being taken seriously within science itself. Indeed, questions about the possibility of consciousness in AI systems – and, indeed, in artificially created biological systems (Box: Consciousness in cerebral organoids?) – are attracting the attention of ethicists, policymakers, and legal theorists.

Box: Consciousness in cerebral organoids?

Derived from stem cells, cerebral organoids are laboratory-made 3D structures that display certain features of the developing human brain, such as spiking activity and complex oscillatory waves (Sakaguchi et al. 2019; Trujillo et al. 2018). Although current cerebral organoids lack many of the neurophysiological features of human brains, the neurofunctional profile of future organoids is likely to increasingly approximate that of the developing human brain.

Because organoids have neither sensory nor motor systems, standard tests for the presence of consciousness cannot be applied to them. One possible solution to this problem involves the redeployment of measures of neural complexity that have been used to identify consciousness in brain-damaged patients. The leading example of such measures is the *perturbational complexity index* (PCI), in which the cortex is first stimulated by transcranial magnetic stimulation (TMS) and then electroencephalography (EEG) is used to measure the complexity of the resulting neural response (Casali et al. 2013). PCI has proven effective in detecting consciousness in a variety of contexts in which the capacity for verbal report and voluntary behaviour is compromised, such as dreaming, ketamine-induced anaesthesia, and severe brain injury (Sarasso et al. 2015; Casarotto et al. 2016), and in principle it might be used to identify consciousness in organoids.

Further reading

Bayne, T., A. Seth, and M. Massimini. 2020. Are There Islands of Awareness? *Trends in Neurosciences*, 43(1): 6–16.

The challenge of artificial consciousness is often approached via the Turing test. Turing, of course, proposed his test as a test for thought, but we can consider its plausibility as a test for consciousness. (Indeed, if thought requires consciousness – as some theorists hold – then the Turing test is a test for thought only if it is also a test for consciousness.) As we noted in Chapter 5, the Turing test involves three individuals: an interrogator, an ordinary human, and an AI system ('the target'). Let's call our target 'Ava', in honour of the protagonist at the centre of *Ex Machina*. Ava and the human are assigned to separate rooms, and Ava can be said to have passed the Turing test if, after an extended session of questioning, the interrogator is unable to tell which room contains the human and which contains her.

Failing the Turing test surely tells us nothing about whether or not Ava is conscious. In the same way that a non-human animal might be conscious but be unable to pass the Turing test, so too we should allow that an artificially conscious agent's mode of consciousness is so unlike ours that its identity would be easily unmasked by the interrogator. But what if Ava *passes* the Turing test? Would that justify the conclusion that she was conscious?

The issue is an open one. What we can say, however, is neither of the two arguments that we considered in connection with the sceptical problem of other minds – the argument from analogy and the inference to the best explanation – is all that promising. The problem with the argument from analogy is there is little reason to assume that the correlations between cognitive/behavioural capacities and consciousness that apply to human beings (and, to some degree perhaps, other species) will apply also to AI systems such as Ava. Of course, if we knew that Ava's cognitive architecture mirrored our own at every level of description then we would be on firmer ground in thinking that she was conscious, but it is surely unlikely that Ava's cognitive architecture will mirror our own at every level of description. Ava is likely to resemble us at some levels of description and to be very different from us at other levels of description. The argument from analogy doesn't tell us which levels of description matter with respect to the question of consciousness. Appeals to inference to the best explanation are equally problematic, for the explanatory role of consciousness in us – let alone in artificial agents – remains obscure. Would an AI system need to be conscious in order to pass the Turing test, or could it display the conversational agility of an ordinary human without possessing even a whiff of subjectivity? At this point, we simply don't know.

In fact, questions about the distribution of consciousness in artificial systems might be beyond our capacity to answer. Here is an argument for precisely that conclusion:

1 A validated test for artificial consciousness must either compare the target AI system to us, or it must appeal to a reductive theory of consciousness.
2 Tests of consciousness that compare the target AI system to us are unfairly parochial.
3 Developing a reductive theory of consciousness requires closing the explanatory gap.
4 We can't close the explanatory gap.

So,

C We can't develop a validated test for artificial consciousness.

Let's call the conclusion of this argument the 'pessimistic inference'. How might it be challenged? As I see it, premise (1) is secure. Premise (2) is also compelling, for it seems likely that most forms of artificial consciousness will differ greatly from the mode of consciousness that

we enjoy; indeed, it seems likely that they will differ greatly from any biologically based mode of consciousness. The controversial premises are (3) and (4), although they will be challenged in different ways by different theorists. These issues are best explored by returning to the contrast between Type-A physicalists and Type-B physicalists (see Chapter 8).

The Type-A physicalist may well grant (3), but they are likely to challenge (4). As they see it, identifying the distribution of consciousness in artificial systems is possible, for there are certain kinds of physical/functional states that necessitate consciousness, and progress in philosophy and/or science will unmask those necessitation relations, thus closing the explanatory gap. And once the explanatory gap is closed, we will be able to tell whether a target AI system is conscious by identifying its physical/functional profile. Of course, whether the Type-A theorist is right to assume that we have the capacity to close the explanatory gap is a contested matter.

The Type-B physicalist is likely to grant (4) but will challenge (3), arguing that it is possible to provide an account of consciousness without fully closing the explanatory gap. The problem, however, is that theories of consciousness are developed by studying consciousness as it occurs in human beings (and, to a lesser extent, other mammals). Thus, any reason to think that a particular theory can be applied to systems that are very different to us must, it seems, diminish in direct proportion to the cognitive gulf between us and the target system. In a nutshell, artificial consciousness confronts us with the following challenge. The more an artificial system differs from us, the less appropriate it is to subject it to tests of human consciousness, but the less a test for consciousness is tailored to us, the more difficult it is to validate. The pessimistic inference is not unassailable, but it should certainly give pause to anyone who is tempted to treat the problem of artificial consciousness with insouciance.

11.6 Conclusion

We began this chapter by considering the three main accounts of the mechanics of mindreading: the perceptual theory, the simulationist theory, and the theory-theory. We saw that although perception might make *a* contribution to mindreading, much of our everyday understanding of the minds of those around us is likely to be done either by simulation-based processes or by the deployment of a rich body of information about how minds work.

We then turned to more distinctively philosophical terrain, beginning with the question of what our psychological concepts must be like in order for them to be applicable to other people. We considered Wittgenstein's argument for thinking that our capacity to intelligibly apply psychological concepts to other people shows that such concepts are not grasped on the basis of attending to 'private' sensations. I suggested that that argument fails: psychological concepts can be coherently applied to other beings even if they are acquired by attending to the character of one's own experiences.

In the second half of this chapter, we turned our attention to the epistemic challenges raised by other minds. We began with the problem of other human minds, and the sceptic's claim that we cannot – or at any rate, do not – know what other people are thinking and feeling. We then considered two ways in which claims to knowledge of other minds might be justified – the analogical response and the inference to the best explanation response – noting some of the main

challenges facing each account. Finally, we explored the question of consciousness in non-human animals and artificial systems. It is here that the weaknesses of the analogical approach and the IBE approach are most evident. The analogical approach provides us with little guidance, for it doesn't tell us what kinds of properties are most relevant to the ascription of consciousness. The IBE approach is perhaps more promising, but it requires an account of what consciousness does (and does not) explain, and at present we have no such account (or at least, not one that elicits general agreement). Perhaps progress in science or philosophy will enable us to get a better grip on how consciousness is distributed in the animal kingdom and artificial systems. Another possibility, however, is that there are principled limits on our capacity to identify consciousness in beings that are radically unlike us.

Further reading

Excellent introductions to the psychology of mindreading debate can be found in Suilin Lavelle's *The Social Mind: A Philosophical Introduction* (Routledge, 2018) and Ian Apperly's *Mindreaders: The Cognitive Basis of 'Theory of Mind'* (Psychology Press, 2010). A number of the foundational papers in the debate about the psychology of mindreading can be found in a pair of volumes edited by Martin Davies and Tony Stone: *Folk Psychology: The Theory of Mind Debate* (Wiley-Blackwell, 1995) and *Mental Simulation: Evaluation and Applications* (Wiley-Blackwell, 1995). Important monographs on the topic include Alvin Goldman's *Simulating Minds: The Philosophy, Psychology and Neuroscience of Mindreading* (OUP, 2006); Stephen Stich and Shaun Nichols's *Mindreading* (OUP, 2003); and Gregory Currie and Ian Ravenscroft's *Recreative Minds: Imagination in Philosophy and Psychology* (OUP, 2002). Peter Godfrey-Smith's 'Folk Psychology as a Model', *Philosophers' Imprint*, 5(6): 1–16 (2005) provides an thought-provoking treatment of the debate between the simulation-theory and the theory-theory, while Shannon Spaulding's 'Mindreading beyond Belief: A More Comprehensive Conception of How We Understand Others', *Philosophy Compass*, *13*(11): e12526 (2018) examines aspects of mindreading that have not traditionally received the attention they deserve.

Discussion of the conceptual and sceptical problems of other minds can be found in Anita Avramides's *Other Minds* (Routledge, 2001) and Alec Hyslop's *Other Minds* (Kluwer, 1995). For more on the analogical account, see Alec Hyslop and Frank Jackson, 'The Analogical Inference to Other Minds', *American Philosophical Quarterly*, 9: 168–76 (1972). A recent collection of papers that address both conceptual and empirical issues relating to our knowledge of other minds can be found in *Knowing Other Minds* (OUP, 2019), edited by Anita Avramides and Matthew Parrott.

The literature on the philosophy of animal minds is vast. A good place to begin is with Kristin Andrews's *The Animal Mind: An Introduction to the Philosophy of Animal Cognition* (2nd ed., Routledge, 2020). Two excellent collections of essays on animal thought are *Rational Animals?* (OUP, 2006), edited by Susan Hurley and Matthew Nudds, and *The Philosophy of Animal Minds* (CUP, 2009), edited by Robert Lutz. For an interesting discussion of the mirror test see Alexandra Boyle's 'Mirror Self-Recognition and Self-Identification', *Philosophy and Phenomenological Research*, 97/2: 284-303 (2018). For discussions that focus specifically on animal consciousness, see Andrew Barron and Colin Klein, 'What Insects Can

Tell Us about the Origins of Consciousness', *PNAS*, 113(18): 4900–4908 (2016); Jonathan Birch, 'The Search for Invertebrate Consciousness', *Noûs*, forthcoming; Michael Tye, *Tense Bees and Shell-Shocked Crabs: Are Animals Conscious?* (OUP, 2017); and Simona Ginsberg and Eva Jablonka, *The Evolution of the Sensitive Soul: Learning and the Origins of Consciousness* (MIT Press, 2019). Also highly recommended is David Foster Wallace's 'Consider the Lobster', in *Consider the Lobster and Other Essays* (Little, Brown, 2005).

Ned Block's 'The Harder Problem of Consciousness', *Journal of Philosophy*, 99(8): pp. 391–425 (2002) explores some of the challenges raised by artificial consciousness. For a thought-provoking treatment of the possibility of love with an AI system, see Spike Jonze's film *Her* and the related discussion in Troy Jollimore's '"This Endless Space between the Words": The Limits of Love in Spike Jonze's *Her*', *Midwest Studies in Philosophy*, 39(1): 120–43 (2015).

Study questions

1 This chapter distinguishes three elements of mindreading. What are they, and how are they related to each other?
2 What are the three accounts of the mechanics of mindreading presented in this chapter? What do you regard as the main strengths and weaknesses of each account?
3 What is the conceptual problem of other minds? Why is it more fundamental than the sceptical (epistemic) problem of other minds? Which of the two solutions to it that are presented in Section 11.2 do you prefer, and why?
4 This chapter distinguishes two conceptions of the sceptical problem of other minds. What is the difference between these two conceptions, and which do you think is more compelling?
5 What is the analogical response to the sceptical problem of other minds? What are its strengths and weaknesses?
6 What is the inference to the best explanation response to the sceptical problem of other minds? What are its strengths and weaknesses?
7 In what sense are the challenges posed by other kinds of minds similar to those that are posed by other human minds, and in what sense are they distinctive? In answering this question, you might like to reflect on the problems posed by ascribing mental states to non-linguistic human beings, such as babies or severely brain-damaged patients.
8 What is the pessimistic inference as outlined in Section 11.5? Do you find the argument for it plausible? If not, how would you challenge it?

Notes

1 Another approach to mindreading, which we won't discuss here, is given by Dennett's intentional stance. See Chapter 6 for discussion.
2 For discussion of perceptual accounts of mindreading, see Carruthers (2015), Cassam (2007), Dretske (1973), Green (2010), Lavelle (2012), McDowell (1998), McNeill (2012), and J. Smith (2015).
3 See Currie and Ravenscroft (2002), Gallese (2005), Goldman (1989, 2006), Gordon (1986), Harris (1992), and Heal (1986).
4 The use of simulation does require that one make certain theoretical assumptions (e.g., in determining what counts as subjecting a model to hurricane-like conditions), but these assumptions don't

drive simulation-based mindreading in the way that they drive theory-based mindreading. See Goldman (1989) for further discussion.

5 For discussion, see Avramides (2001), Gomes (2011), McGinn (1984), and Nagel (1986).

6 Curiously, discussions of the sceptical problem of other minds rarely engage with the psychology of mindreading, and it is sometimes tempting to read certain authors as concerned with the question of whether we could in principle have knowledge of other minds as opposed to engaging with the question of whether our ordinary mindreading practices do in fact deliver knowledge.

7 Although J. S. Mill is associated with the analogical account, Thomas (2001) argues convincingly that Mill's view was actually closer to the inference to the best explanation account that we consider next.

8 Indeed, there is some reason to think that qualia inversion might actually occur. See Nida-Rümelin (1996b) for discussion.

9 For more on octopus consciousness, see Godfrey-Smith (2013, 2016), Mather (2008, 2019), and Seth (2016).

10 For elephants, see Plotnik et al. (2006); for dolphins, see Mitchell (1995) and (Reiss & Marino 2001); for magpies, see Prior et al. (2008); and for fish, see Kohda et al. (2019).

11 For more on the debate between these two interpretations, see Suddendorf and Busby (2003), Suddendorf and Corballis (2007), Clayton et al. (2003a, 2003b).

12 See Boyle (2020-a) for an interesting solution to this challenge.